Mahavir Education Trust's
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Plagiarism Detection Engine Using Binary Search Trees

Submitted By:

- Jaydeep H. Labor | Roll No.: 29
- Swapnil | Roll No.: 24
- Aditya Kharat | Roll No.: 27

Department: Artificial Intelligence & Data Science
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1. INTRODUCTION

Data Structures are an important part of computer science and software development. They provide systematic methods for organizing, storing, accessing, and processing data efficiently.
The objective of this microproject is to apply the concepts of Data Structures & Complexity Analysis to a practical problem. The selected problem is "Plagiarism Detection Engine using Binary Search Trees". The project will involve identifying the problem, studying existing approaches, selecting a suitable data structure, designing an algorithm, and developing a software-based solution.
Stage 1 focuses mainly on problem identification, background study, objectives, feasibility, selection of data structures, and proposed methodology.

2. PROBLEM STATEMENT

Title of the Problem: Plagiarism Detection Engine using Binary Search Trees
Problem Statement: Manual plagiarism checking and standard line-by-line text comparisons are extremely slow and inefficient when scanning large documents. Searching, inserting, and evaluating text records require an organized method of data management to compare student submissions against a reference database quickly. Therefore, a Plagiarism Detection Engine is proposed using Arrays and Binary Search Trees (BST) to perform these operations efficiently.

3. MOTIVATION

The motivation for selecting this project is to understand the practical importance of data structures in solving real-world problems. The project provides students with an opportunity to:

- Apply theoretical concepts of data structures.
- Develop logical and analytical thinking.
- Solve a real-world computational problem.
- Select an appropriate data structure.
- Design efficient algorithms.
- Understand time and space complexity.
- Improve programming and problem-solving skills.

4. AIM OF THE PROJECT

The main aim of the proposed microproject is: "To design and develop an efficient solution for automated text plagiarism detection using appropriate data structures (Arrays and Binary Search Trees) and algorithms."

5. OBJECTIVES

- To understand and analyze the selected problem.
- To study existing approaches for solving the problem.
- To identify suitable data structures.
- To design an appropriate algorithm.
- To develop an efficient computational solution.
- To analyze the time and space complexity.
- To test the proposed solution using suitable test cases.
- To develop practical programming and problem-solving skills.

6. SCOPE OF THE PROJECT

The proposed project will focus on developing a basic working solution for the identified problem. The scope includes: Tokenizing input text files using Arrays, inserting reference text into a Binary Search Tree, searching suspect phrases, and calculating a percentage similarity score. The project may be further extended in future by incorporating advanced data structures, database connectivity, graphical interfaces, or web/mobile functionality.

7. BACKGROUND STUDY / LITERATURE SURVEY

A preliminary study of the selected problem and related data structures was conducted to understand existing solutions and identify an appropriate approach.

Sr. No.	Approach Studied	Key Observation
1	Manual Checking	Very slow and prone to human error.
2	Linear Array Search	High time complexity of O(N x M) on large datasets.
3	Hash Tables	Fast lookup but loses relative ordering of

Sr. No.	Approach Studied	Key Observation
		sentences.
4	Binary Search Trees	Enables efficient logarithmic O(log N) search times while maintaining structure.

Summary of Literature Survey: The background study indicates that appropriate selection of data structures can significantly improve the efficiency of computational operations. Based on the study, the proposed project will use **Binary Search Trees and Arrays** because it is suitable for **fast, organized hierarchical text searching and token preprocessing**.

8. EXISTING SYSTEM

Existing Approach: Manual checking or basic brute-force string matching without data indexing.
Limitations of Existing Approach: High latency, extreme time consumption on large documents, and lack of automated efficiency.

9. PROPOSED SYSTEM

The proposed system aims to overcome the identified limitations by implementing an organized data structure-based solution.
Proposed Approach: A software-based text comparison tool utilizing Arrays for text tokenization and Binary Search Trees (BST) for fast reference lookups. The proposed system will provide functionalities such as: file input, array preprocessing, BST node insertion, tree searching, and similarity score output.

10. SELECTION OF DATA STRUCTURE

Data Structure	Purpose	Reason for Selection
Arrays	Initial text tokenization and	Contiguous memory

Data Structure	Purpose	Reason for Selection
	raw data storage	allocation allows fast sequential file loading.
Binary Search Tree (BST)	Organizing reference chunks and searching matches	Provides efficient O(log N) search time complexity.

11. PROPOSED ALGORITHMS

- **Insertion:** Inserting reference text tokens into the BST.
- **Deletion:** Removing outdated reference nodes.
- **Searching:** Traversing the BST to check for suspect document matches.
- **Sorting:** Utilizing in-order traversal to retrieve sorted text elements.
- **Traversal:** In-order, pre-order, and post-order tree traversals.

12. PROPOSED SYSTEM ARCHITECTURE / FLOW

Flow: Input -> Data Processing -> Data Structure -> Operations -> Output
[Insert preliminary flowchart here showing text input, array parsing, BST building, and similarity matching]

13. FUNCTIONAL REQUIREMENTS

Sr. No.	Functionality	Description
1	File Input	Accept original reference text and suspect text files.

Sr. No.	Functionality	Description
2	Preprocessing	Tokenize text lines using Arrays.
3	BST Insertion	Build reference tree database.
4	Similarity Check	Search BST nodes to find identical text fragments.
5	Report Generation	Output a final percentage similarity score.

14. SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements: Java, VS Code/Eclipse, Windows/Linux, Git.
Hardware Requirements: Intel Core i3 or higher, 4 GB RAM, 500 MB Storage.

15. EXPECTED OUTCOMES

At the end of the project, students are expected to: Develop a working solution, apply suitable data structures and algorithms, perform fundamental operations efficiently, analyze complexity, test/validate the solution, and demonstrate practical application.

16. COURSE OUTCOME MAPPING

Course Outcome	Contribution of Microproject	Level
CO1	Application of fundamental data structure concepts	3
CO2	Implementation of linear data structures (Arrays)	3
CO3	Application of non-linear data structures (BST)	3
CO4	Algorithm development and complexity analysis	4
CO5	Selection of appropriate data structures for problem solving	3
CO6	Design and implementation of practical solution	3

17. PROJECT WORK PLAN

Stage	Activity	Proposed Timeline
Stage 1	Problem identification and literature survey	Weeks 1-2
Stage 1	Data structure and algorithm selection	Weeks 3-4
Stage 2	Algorithm design and coding	Weeks 5-7
Stage 2	Testing and debugging	Week 8
Stage 3	Results and analysis	Week 9
Stage 3	Documentation and presentation	Week 10

18. EXPECTED COMPLEXITY ANALYSIS

Operation	Expected Time Complexity	Expected Space Complexity
Insertion	O(log N)	O(N)
Deletion	O(log N)	O(N)
Searching	O(log N)	O(1)
Sorting	O(N)	O(N)
Traversal	O(N)	O(H)

19. CONCLUSION OF STAGE 1

Stage 1 of the microproject focused on understanding and analyzing the selected problem, studying existing approaches, identifying the limitations, and proposing an appropriate data structure-based solution. The next stage will focus on detailed algorithm development, program implementation, testing, debugging, and performance analysis.

20. REFERENCES

- Reema Thareja, Data Structures Using C, Oxford University Press
- Ellis Horowitz, Sartaj Sahni and Susan Anderson-Freed, Fundamentals of Data Structures
- Thomas H. Cormen et al., Introduction to Algorithms

FACULTY EVALUATION – STAGE 1

Parameter	Marks
Problem Identification	____ / 10
Problem Statement	____ / 10
Motivation and Objectives	____ / 10
Literature Survey	____ / 15
Data Structure Selection	____ / 15
Proposed Methodology	____ / 15
Flowchart / System Design	____ / 10
Presentation / Viva	____ / 15
Total	**____ / 100**